
Distractor Filtering Through Open Knowledge Graph Concept Proximity for Dense RAG Systems

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Abstract

Dense retrieval can surface *distractors*: passages semantically close to the query that do not contain the answer, degrading RAG accuracy (Cuconasu et al., 2024). We test whether *pure graph topology* on Wikidata can filter them, without using relation semantics. We build a 661.5M-edge entity graph from a local HDT dump, score query–document pairs by topological reachability ($\text{connected} \times \text{purity}$), and fuse this with the dense score across a grid of hop-distance, hub-threshold and mixing weight α , evaluated on 1000 NQ-open queries with an LLM judge. The result is a clean *negative*: the KG signal never beats the dense baseline (0.364) beyond noise (best 0.372) and actively degrades accuracy as its weight grows (down to 0.290); 13/90 conditions are significant under McNemar+Bonferroni, all of them worse. We explain the mechanism (reachability saturates on a dense graph) and document the engineering and the methodological departures from (Cuconasu et al., 2024).

1. Introduction

Retrieval-Augmented Generation (RAG) grounds LLMs (Naveed et al., 2023) in external text, but (Cuconasu et al., 2024) show that high-scoring retrieved passages are often *distractors* — semantically similar to the query yet answer-free — and hurt the reader. We ask whether the topology of an open knowledge graph (Wikidata) can separate relevant passages from distractors. The hypothesis: a passage whose entities are *graph-close* to the query entities is less likely to be a distractor than one chosen by semantic similarity alone. Unlike approaches that exploit the graph’s full relation semantics, we test the weakest signal — bare proximity within k hops — to

isolate its contribution.

Our contributions are: (i) a pipeline that navigates the *entire* Wikidata graph from a local HDT dump and tests ≤ 3 -hop reachability meet-in-the-middle, without ever expanding a high-degree hub’s neighbour list; (ii) a clean negative result, supported by exact McNemar tests and a mechanistic explanation; (iii) three deliberate deviations from the evaluation setup of (Cuconasu et al., 2024) (passage padding, an extraction prompt for the base model, an LLM judge), documented for reproducibility. Code: <https://github.com/<USER>/dl-RAG-denseAndKG>.

2. Related work

We adopt the corpus, retriever and reader of (Cuconasu et al., 2024), who characterise the distractor problem in RAG. Retrieval uses Contriever over the DPR Wikipedia corpus; entity linking uses ReFinED (Ayoola et al., 2022), one of the end-to-end systems evaluated by (Bast et al., 2023). Knowledge-graph approaches to RAG use the graph’s relations and semantics; we instead use only its topology. The graph is stored as HDT over a Wikidata dump.

3. Method

Data and retrieval. The corpus is the Dec-2018 Wikipedia release of (Cuconasu et al., 2024), Contriever-encoded and indexed with FAISS (IndexFlatIP, ~ 42 M passages). From NQ-open we keep queries with ≤ 5 -token answers where the question and *all* answer variants link to Wikidata entities (ReFinED), and evaluate on a curated 1000-query subset.

Passage segmentation (departure 1). DPR/(Cuconasu et al., 2024) use raw 100-word passages with ragged, unpadded tails. We instead cut on sentence boundaries and pad each segment back to exactly 100 words using the article’s opening words. This avoids subject-less chunks (e.g. a passage starting mid-sentence) that break entity linking and reader comprehension, keeps passage length uniform, and incidentally anchors every passage to the article sub-

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ject. Contriever mean-pools, so fixed length is a consistency choice, not a model requirement.

Graph construction (engineering). We query Wikidata from a local HDT dump (Jan 2022, ~ 166 GB) via pyHDT. The public SPARQL endpoint was abandoned after even minimal COUNT queries on hubs (e.g. Q5, *human*) timed out. We export all entity-to-entity `wdt : *` triples per predicate, yielding 6.61×10^8 edges. For a pair (q, d) we decide $\text{dist}(q, d) \leq 3$ meet-in-the-middle: distance 1 is a set lookup, distance 2 an intersection of 1-hop neighbourhoods N_1 , distance 3 an indexed edge-probe between $N_1(q)$ and $N_1(d)$. We never materialise the neighbourhood of a hub (Q30, *USA*, has 2.35M edges); we only test membership. A degree threshold t optionally drops high-degree bridge nodes.

Scoring and fusion. Following the proposal, `kg_score = connected_ratio × purity_ratio`, where `connected_ratio` is the fraction of query entities reaching ≥ 1 document entity within k hops and `purity_ratio` the fraction of document entities reached by any query entity. We fuse with the dense score as

$$\text{final} = (1 - \alpha) \text{dense_norm} + \alpha \text{kg_score}, \quad (1)$$

with α the *KG weight* and `dense_norm` a per-query max-only rescaling ($s / \max_q s$) that caps the dense score at 1 without stretching its narrow band (min-max would). We sweep a grid of distance $k \in \{1, 2, 3\}$, threshold $t \in \{500, 1k, 2k, 5k, 10k, \infty\}$ and $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$: 90 rerank cells plus the dense baseline.

Generation and evaluation (departures 2–3). The top-5 passages feed Llama-2-7B (base). The prompt is derived from (Cuconasu et al., 2024) with two deviations: a one-shot exemplar (the base, non-instruct model otherwise continues “Answer:” with Wikipedia-style prose) and the question placed *above* the documents (the paper places it after). Correctness is decided by an *LLM judge* (Qwen2.5-7B-Instruct, a different model to avoid self-preference) that checks whether the response contains a gold answer. This replaces the brittle substring exact-match of (Cuconasu et al., 2024), which misses paraphrases, formatting and preambles.

4. Results

Baseline is sound. Dense retrieval scores 0.364, in line with the ≈ 0.36 that (Cuconasu et al., 2024) report for Llama-2 with Contriever-retrieved documents on NQ-open — the harness is healthy and the comparison valid.

No gain, then harm. The best cell ($\alpha=0.3$, dist 2, $t=5k$) reaches 0.372: $+0.8pp \approx 8$ queries, i.e. noise (Table 1).

Table 1. LLM-judge accuracy, all 90 rerank settings (1000 NQ-open queries). Dense baseline = 0.364. Columns: KG weight α ; rows: hop-distance k , hub-threshold t (∞ =no filter; distance 1 has no bridge, so it is threshold-invariant). Best **bold**, worst underlined.

k	t	$\alpha=.1$	.3	.5	.7	.9
1	any	.353	.344	.343	.316	.322
2	500	.362	.358	.361	.348	.354
2	1,000	.364	.361	.358	.343	.350
2	2,000	.360	.371	.357	.337	.341
2	5,000	.357	.372	.362	.337	.337
2	10,000	.364	.365	.357	.334	.329
2	∞	.354	.343	.311	.298	.295
3	500	.354	.344	.341	.329	.335
3	1,000	.346	.355	.342	.322	.310
3	2,000	.354	.349	.316	.303	.293
3	5,000	.366	.351	.310	.292	<u>.290</u>
3	10,000	.361	.362	.321	.306	.304
3	∞	.347	.333	.314	.314	.318

As α grows the KG signal dominates and accuracy falls monotonically, bottoming at 0.290 ($\alpha=0.9$, dist 3, $t=5k$).

Threshold gradient is the mechanism. Holding $\alpha=0.9$, dist 2, accuracy decays monotonically as the hub filter loosens, from 0.354 ($t=500$) to 0.295 ($t=\infty$): admitting high-degree nodes as bridges connects everything to everything and destroys the signal. (At dist 1 the threshold is inert — all cells coincide, a passed sanity check.)

Significance. Exact McNemar tests against the baseline, Bonferroni-corrected for the 90 comparisons (significance threshold $0.05/90 \approx 5.6 \times 10^{-4}$), flag 13 of the 90 conditions as significant — *all* worse, none better; each flips $\approx 200/1000$ answers yet nets negative.

5. Discussion and conclusions

Why it fails. On a graph as dense as Wikidata, 2–3-hop reachability *saturates*: almost every document entity is reachable from almost every query entity, so `connected_ratio` ≈ 1 and the score stops discriminating; the hub filter limits the damage but recovers no signal. For about 17% of queries the question entity is itself a very high-degree node (a country, a religion), where any passage is graph-close and the KG score is uninformative.

Limitations and future work. Binary reachability collapses relation semantics (P31, *instance-of*, and P19, *born-in*, count alike) and the set-overlap metric ignores edge weights. The honest next step is to use the graph’s relations and semantics — not just its bare topology.

Conclusion. Pure topological proximity does not filter distractors for dense RAG on NQ-open; the value is the explained negative result and the reusable Wikidata tooling.

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